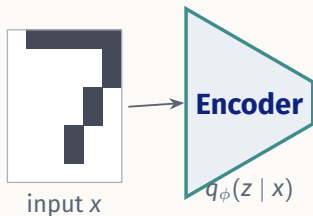


Encoder Output: μ and σ

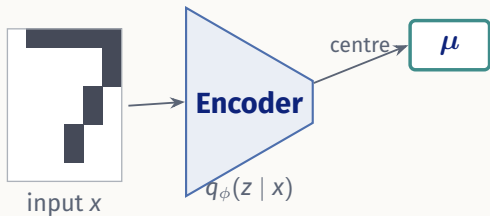


Encoded distribution for one input

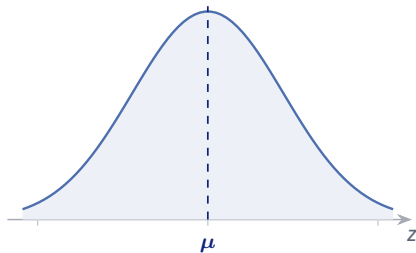


The encoder reads the input and extracts the information needed for its latent representation.

Encoder Output: μ and σ

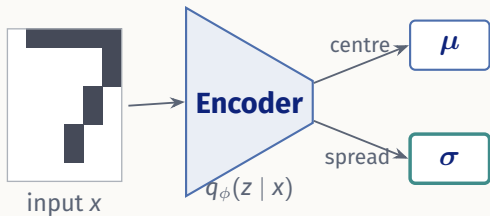


Encoded distribution for one input

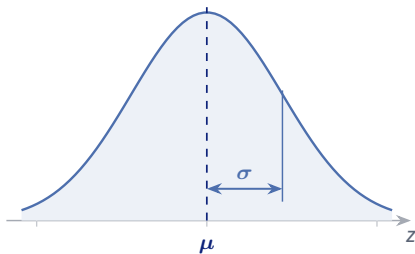


The mean μ locates the centre of the encoded example in latent space.

Encoder Output: μ and σ

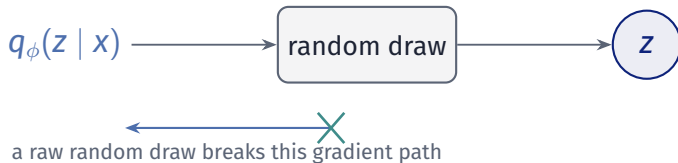


Encoded distribution for one input



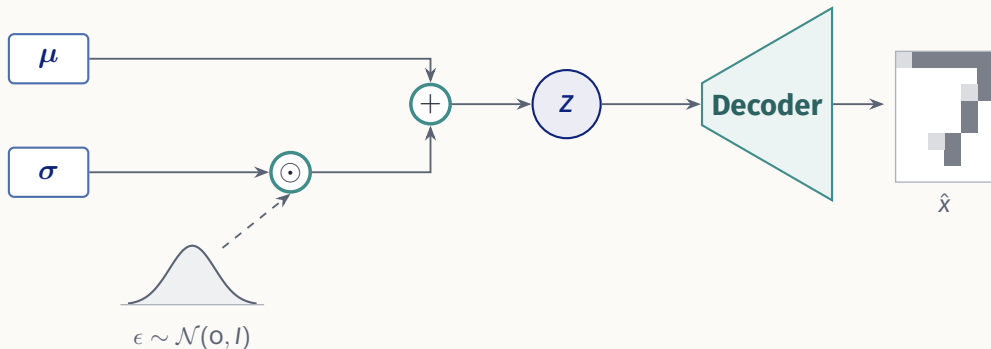
The standard deviation σ expresses uncertainty around that centre: $q_{\phi}(z | x) = \mathcal{N}(\mu, \sigma^2)$.

Reparameterised Sampling



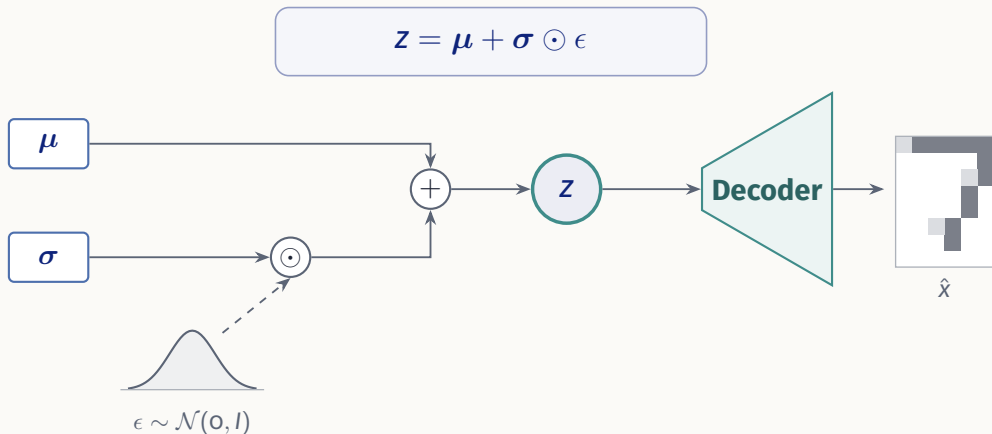
Training needs a differentiable route through the latent sample.

Reparameterised Sampling



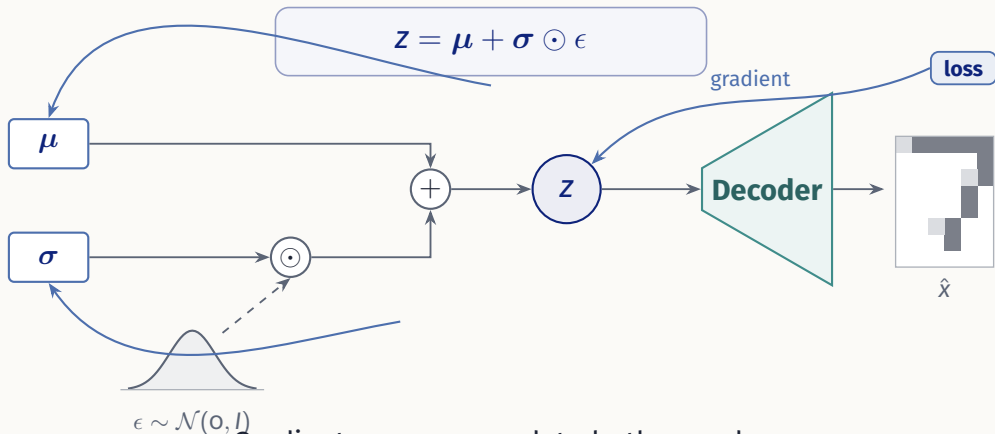
The reparameterisation trick isolates the randomness in ϵ .

Reparameterised Sampling



The sample stays random, while z becomes an explicit function of μ and σ .

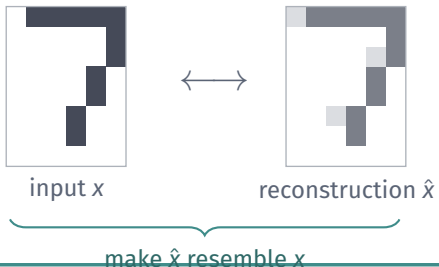
Reparameterised Sampling



Gradients can now update both encoder parameters through the sampled value of z .

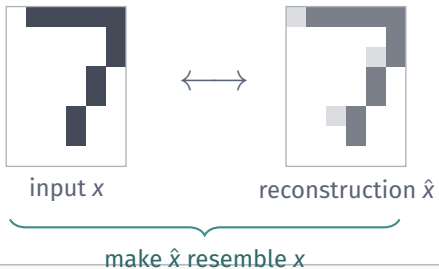
Reconstruction and KL Loss

Reconstruction loss

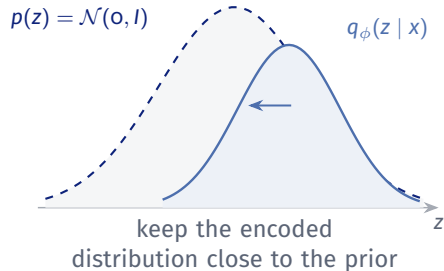


Reconstruction and KL Loss

Reconstruction loss

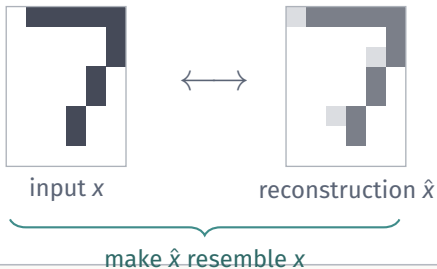


KL regularisation

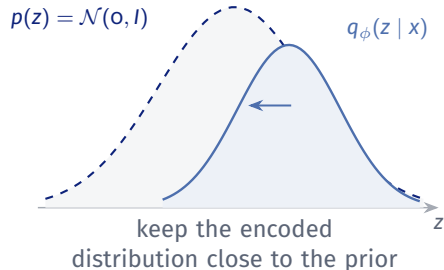


Reconstruction and KL Loss

Reconstruction loss



KL regularisation

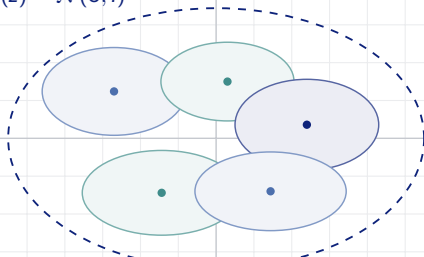


$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathcal{L}_{\text{recon}}(x, \hat{x})}_{\text{preserve the input}} + \underbrace{D_{\text{KL}}(q_\phi(z | x) || p(z))}_{\text{organise latent space}}$$

A Smooth, Sampleable Latent Space

z_2 After training

$$p(z) = \mathcal{N}(0, I)$$



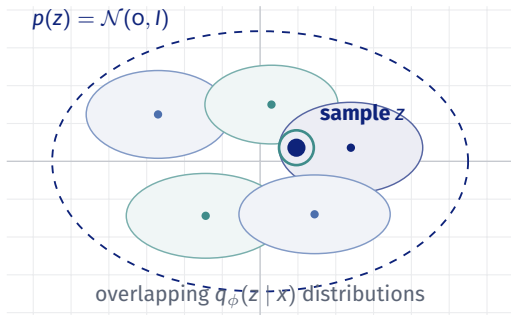
overlapping $q_\phi(z | x)$ distributions

z_1

A Smooth, Sampleable Latent Space

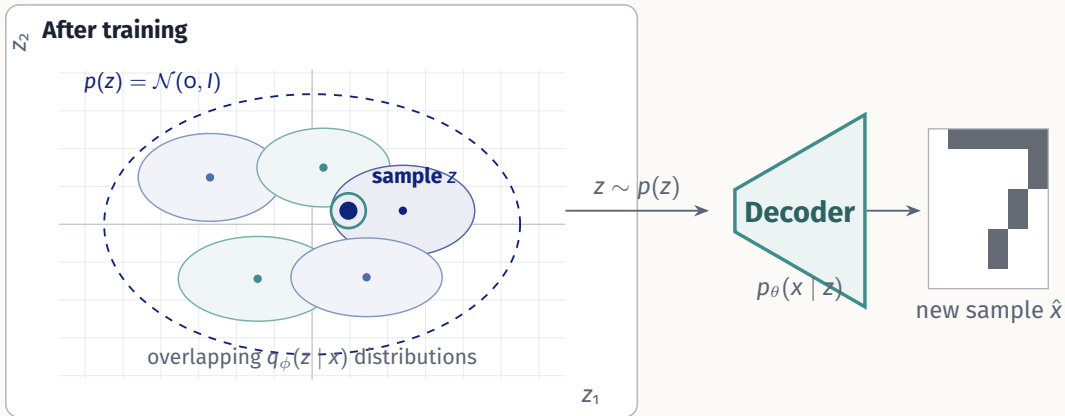
z_2 After training

$$p(z) = \mathcal{N}(0, I)$$



z_1

A Smooth, Sampleable Latent Space



A Smooth, Sampleable Latent Space

